

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series

Topics: Geometric Series • Power Series • Maclaurin Series • Taylor Series • Error Analysis

Engineering Scenario: Sloping structural system vertical component $y = L \sin(\theta)$ (where $L = 20$ m)

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PART 1: GEOMETRIC SERIES

Investigation Question: Compare partial sums with the exact value for $x = 0.5, 0.8$, and 0.9 . How does the number of terms affect accuracy? Why does convergence speed differ?

ANSWER & INVESTIGATION RESULTS

Using the geometric series sum function $S_N = 1 + x + x^2 + \dots + x^N$, we calculate the partial sums and absolute/percentage errors against the exact closed-form value $1 / (1 - x)$:

x	Exact $1/(1-x)$	N = 3 (Error)	N = 5 (Error)	N = 10 (Error)	N = 20 (Error)
0.5	2.00000	1.87500 (6.25%)	1.96875 (1.56%)	1.99902 (0.05%)	1.99999 (<0.001%)
0.8	5.00000	2.95200 (40.96%)	3.68928 (26.21%)	4.58053 (8.39%)	4.94235 (1.15%)
0.9	10.00000	3.43900 (65.61%)	4.68559 (53.14%)	6.86189 (31.38%)	8.78423 (12.16%)

Key Concepts & Findings:

- Effect of Number of Terms:** Increasing the number of terms N systematically reduces the truncation error for all $|x| < 1$.
- Convergence Speed Variance:** Convergence speed depends directly on the ratio x^k . For $x = 0.5$, the terms $(0.5)^k$ decay exponentially fast, reaching high accuracy with few terms. For $x = 0.9$, $(0.9)^k$ decays very slowly, requiring over 50+ terms to achieve comparable precision.

PART 2: POWER SERIES

Task & Concept: Generalize the geometric series into a power series $P_N(x) = a_0 + a_1x + a_2x^2 + \dots + a_Nx^N$. How do power series relate to transcendental functions?

ANSWER & KEY INSIGHT

A polynomial is a finite power series. An infinite power series allows us to represent smooth transcendental functions (such as $\sin(\theta)$, $\cos(\theta)$, and e^x) as continuous polynomials. In engineering computing, this enables complex

transcendental calculations to be performed using simple additions and multiplications inside CPU arithmetic logic units (ALUs).

PART 3: MACLAURIN SERIES FOR SIN(Θ)

Investigation Question: For $\theta = 10^\circ$ (0.174533 rad), calculate approximations using 1, 2, 3, and 4 terms. Compare each to the exact value. What is the error at each step?

ANSWER & NUMERICAL EVALUATION

Exact value: $\sin(10^\circ) = \sin(0.174532925 \text{ rad}) = 0.173648178$

Terms (N)	Series Expansion Formula	Approximation Value	Absolute Error	Percentage Error
1 Term	θ	0.174532925	8.8475×10^{-4}	0.50950%
2 Terms	$\theta - \theta^3/6$	0.173646828	1.3497×10^{-6}	0.00078%
3 Terms	$\theta - \theta^3/6 + \theta^5/120$	0.173648179	1.0221×10^{-9}	0.00000059%
4 Terms	$\theta - \theta^3/6 + \theta^5/120 - \theta^7/5040$	0.173648178	$< 1.0 \times 10^{-12}$	$< 1.0 \times 10^{-10}\%$

PART 4: ENGINEERING INVESTIGATION (Y = L SIN(Θ), L = 20 M)

- Analysis Questions:
- How does the error change as the angle increases?
 - How does the error change as you add more terms?
 - For small angles ($< 5^\circ$), how many terms are sufficient?
 - For larger angles ($> 20^\circ$), how many terms are needed?

ANSWER & NUMERICAL ANALYSIS

Calculated Table for Maclaurin Approximations of Vertical Length y (Exact vs. N Terms):

Angle (θ)	Exact y (m)	N=1 Approx (Error %)	N=2 Approx (Error %)	N=3 Approx (Error %)
1°	0.349048	0.349066 (0.0051%)	0.349048 (<0.0001%)	0.349048 (<0.0001%)
2°	0.697989	0.698132 (0.0203%)	0.697989 (<0.0001%)	0.697989 (<0.0001%)
5°	1.743115	1.745329 (0.1270%)	1.743113 (0.0001%)	1.743115 (<0.0001%)
10°	3.472964	3.490659 (0.5095%)	3.472937 (0.0008%)	3.472964 (<0.0001%)
15°	5.176381	5.235988 (1.1515%)	5.176003 (0.0073%)	5.176382 (<0.0001%)
20°	6.840403	6.981317 (2.0600%)	6.838271 (0.0312%)	6.840416 (0.0002%)
30°	10.000000	10.471976 (4.7198%)	9.682332 (0.3176%)	10.004402 (0.0440%)

Answers to Specific Questions:

- Error Growth vs. Angle:** Absolute and percentage errors increase significantly as the angle increases. Higher power terms (θ^3, θ^5) grow larger rapidly with increasing θ .
- Error Reduction vs. Terms:** Adding terms dramatically reduces error by orders of magnitude due to the fast growth of factorials ($(2n+1)!$) in the denominators.
- Small Angles ($< 5^\circ$):** 1 term ($y \approx L\theta$) is sufficient, maintaining error well below 0.13% (and below 0.05% for $\theta \leq 3^\circ$).
- Larger Angles ($> 20^\circ$):** At least 3 terms are required to guarantee less than 0.1% percentage error at 30° (where 2 terms yields 0.318% error, but 3 terms yields 0.044%).

PART 5: TAYLOR SERIES

Comparison Question: For the same angles (1°–30°), calculate errors using Maclaurin (centered at 0°) and Taylor (centered at 10°). Which performs better near 10°? Which is better far from 10°?

ANSWER & COMPARATIVE ANALYSIS (N = 2 TERMS)

Angle (θ)	Exact y (m)	Maclaurin ($a=0^\circ$, N=2) Error	Taylor ($a=10^\circ$, N=2) Error	Better Performing Expansion
1°	0.349048	< 0.0001%	0.3751%	Maclaurin ($a=0^\circ$)
5°	1.743115	0.0001%	0.0421%	Maclaurin ($a=0^\circ$)
10°	3.472964	0.0008%	0.0000%	Taylor ($a=10^\circ$)
15°	5.176381	0.0073%	0.0336%	Maclaurin ($a=0^\circ$)
20°	6.840403	0.0312%	0.1205%	Maclaurin ($a=0^\circ$)
30°	10.000000	0.3176%	0.3951%	Maclaurin ($a=0^\circ$)

Findings:

- Near 10°:** The Taylor series centered at $a = 10^\circ$ performs significantly better near 10° (achieving exact matching at 10°).

- **Far from 10° :** Maclaurin performs better for angles close to 0° (1° – 5°), whereas Taylor centered at 10° loses accuracy as the evaluation point moves further from its center a .

PART 6: ENGINEERING DECISION — ERROR TOLERANCE

Task 1: Minimum terms for $< 0.1\%$ tolerance (1° – 30°) for Maclaurin vs. Taylor ($a=10^\circ$).

Task 2 & Critical Angle: At what angle does $\sin(\theta) \approx \theta$ stop being sufficiently accurate ($< 0.1\%$ error)?

Analysis Questions 1–5:

1. Does adding more terms always improve the approximation?
2. Why is the Maclaurin approximation most accurate near zero?
3. Why can changing the Taylor expansion point improve accuracy?
4. How does the distance from the expansion point affect accuracy?
5. What is the critical angle for the $\sin(\theta) \approx \theta$ approximation?

ANSWER & TOLERANCE ANALYSIS

Task 1: Minimum Terms Needed to Achieve $< 0.1\%$ Error

Angle Range	Maclaurin Min Terms ($a=0^\circ$)	Taylor Min Terms ($a=10^\circ$)
1° to 4°	1 Term ($y = L\theta$)	2 Terms
5° to 18°	2 Terms ($y = L(\theta - \theta^3/6)$)	2 Terms
19° to 30°	3 Terms ($y = L(\theta - \theta^3/6 + \theta^5/120)$)	3 Terms

Task 2 & Critical Angle Answer:

The 1-term small-angle approximation $\sin(\theta) \approx \theta$ reaches exactly 0.1% error at $\theta = 4.45^\circ$ (**0.0777 rad**). Beyond **4.45°** , it stops being sufficiently accurate for a 0.1% tolerance requirement.

Answers to Analysis Questions:

1. **Does adding terms always improve accuracy?** Yes. Because the Taylor/Maclaurin series for $\sin(\theta)$ converges globally for all real numbers, adding terms strictly decreases the truncation error.
2. **Why is Maclaurin most accurate near zero?** Maclaurin is centered at $a = 0$. The deviation term $(\theta - 0)^n = \theta^n$ is smallest near zero.
3. **Why change expansion point?** Centering at $a \neq 0$ makes $(\theta - a)$ small for angles near a , transferring the zone of maximum precision to that operational region.
4. **Distance effect:** Truncation error grows rapidly as $|\theta - a|$ increases, because $(\theta - a)^n$ increases power-wise.
5. **Critical Angle:** 4.45° .

PART 7: FINAL ENGINEERING RECOMMENDATION

Decision Problem: You are the civil engineer responsible for the calculation requiring $< 0.1\%$ error. Would you use Maclaurin series, Taylor series centered closer to the angle of interest, or the exact sine function? Defend your choice.

OFFICIAL ENGINEERING DECISION & DEFENSE

Primary Recommendation (Digital Solvers & Software):

Use the **exact sine function** ($y = L \sin(\theta)$) via native hardware math libraries like Python's `math.sin`). Modern hardware computes exact trigonometric values using CORDIC algorithms with near-zero latency ($\sim$ nanoseconds) and full double-precision accuracy (10^{-16}), completely eliminating truncation risk.

Secondary Recommendation (Embedded Hardware / Manual Calculations):

If restricted to low-power embedded controllers or manual field calculations:

- **For Angles 0° to 4.45° :** Use **1-term Maclaurin** ($y = L\theta$). Terms: 1 | Error: $< 0.1\%$.
- **For Angles 0° to 30° :** Use **3-term Maclaurin series** ($y = L(\theta - \theta^3/6 + \theta^5/120)$). Terms: 3 | Max Percentage Error at 30° : **0.044%** (well under 0.1%).

Justification of Tradeoffs:

While a Taylor series centered at $a = 10^\circ$ provides excellent localized precision, it requires storing extra constants ($\sin(10^\circ)$, $\cos(10^\circ)$) and introduces extra operations ($\theta - a$). A 3-term Maclaurin series delivers uniform $< 0.05\%$ error across the entire operational range (0° – 30°) with far greater structural simplicity.